COS736 QUESTION 1B RESULTS AND OBSERVATIONS

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Course: Big Data Engineering

Total Emissions per Vehicle Type per Route

Route Type: Inter_City

Vehicle Type	Total Emissions (kgCO2e)
Heavy Truck	3,121,366,803.92
Diesel Van (Euro 4)	624,479.19
Diesel Van (Euro 6)	285,393.68
Electric Van (EV)	72,566.25
TOTAL	3,122,349,243.03

Route Type: Inter_District

Vehicle Type	Total Emissions (kgCO2e)
Diesel Van (Euro 4)	463,124.40
Diesel Van (Euro 6)	222,284.18
Motorcycle (Courier)	70,733.78
Electric Van (EV)	57,503.71
TOTAL	813,646.07

Route Type: Urban_Last_Mile

Vehicle Type	Total Emissions (kgCO2e)
Diesel Van (Euro 4)	12,780.79
Diesel Van (Euro 6)	7,841.65
Drone Delivery	4,873.17
Electric Van (EV)	1,772.59
Cargo Bicycle	0.00
TOTAL	27,268.20

Observation: Heavy Truck recorded the highest emissions across Inter_City routes, with 3,121,366,803.92 kgCO2e, driven by its high base emission factor of 1.2 kg/CO2/km combined with long travel distances exceeding 100km. Cargo Bicycle and Drone Delivery recorded the lowest emissions due to their near-zero base factors.

Routes Ranked by Total Distance Travelled

Rank	Route Type	Total Distance (KM)	Share (%)
1	Inter_City	984,136.70	79.7%
2	Inter_District	217,828.80	17.6%
3	Urban_Last_Mile	32,251.40	2.6%

Observation: Inter_City routes dominate total distance travelled at 79.7% of the total, which is expected as each trip exceeds 100km. Urban Last Mile routes cover the least distance as trips are under 20km, reflecting short last-leg deliveries within city limits.

Eco-Friendly vs Non-Eco-Friendly Total Distance

Category	Vehicles Included	Total Distance (KM)	Percentage
Eco-Friendly	Cargo Bicycle, Drone Delivery, Electric Van (EV)	164,501.50	13.3%
Non-Eco-Friendly	Motorcycle, Diesel Van (Euro 4 & 6), Heavy Truck	1,069,715.40	86.7%

Observation: Non-eco-friendly vehicles covered 86.7% of the total distance, indicating a heavy reliance on fossil-fuel based transport in the logistics network. This presents a significant opportunity to reduce carbon emissions by transitioning more routes to electric or zero-emission vehicles.

Total Emissions per Vehicle Type per Month

Month	Cargo Bicycle	Diesel Van (Euro 4)	Diesel Van (Euro 6)	Drone Delivery	Electric Van (EV)	Heavy Truck	Motorcycle (Courier)
January	0	101,438	45,455	445	11,126	305,128,771	6,700
February	0	91,652	42,564	349	10,701	252,094,335	5,415
March	0	101,925	37,266	451	10,893	216,193,963	6,052
April	0	88,869	45,088	412	12,712	289,930,685	6,408
May	0	88,639	46,806	440	10,370	265,537,711	5,030
June	0	89,978	36,780	381	8,838	233,048,388	5,603
July	0	84,055	48,619	378	10,633	252,524,302	4,906
August	0	92,106	43,313	356	11,103	227,333,654	6,850
September	0	88,062	34,932	333	11,501	268,753,301	5,403
October	0	89,398	49,625	439	12,437	279,469,629	6,805
November	0	83,631	42,040	416	10,453	220,565,034	7,005
December	0	100,630	43,031	473	11,074	310,787,029	4,558

Observation: Heavy Truck consistently produced the highest emissions every month across all vehicle types. December recorded the highest total emissions while March recorded the lowest, suggesting seasonal variation in delivery volumes. Cargo Bicycle and Drone Delivery maintained near-zero emissions throughout the year due to their low base emission factors.